

Python Basics: Variables and Input Function

This notebook covers:

1. Examples of all basic types of variables in Python
2. Printing variables
3. Each and every aspect of the `input()` function

Note: This notebook only uses Python basics. No conditions, loops, or advanced topics are included.

1. Variables in Python

A variable is used to store data in memory.

Python automatically understands the type of data based on the value stored.

Integer Variable (int)

Used to store whole numbers.

In [1]:

```
age = 15  
print(age)
```

Floating Point Variable (float)

Used to store decimal numbers.

In [2]:

```
height = 5.6  
print(height)
```

String Variable (str)

Used to store text. Text must be written inside quotes.

In [3]:

```
name = "Rahul"  
print(name)
```

Boolean Variable (bool)

Used to store True or False values.

In [4]:

```
is_student = True  
print(is_student)
```

Multiple Variables

Python allows storing different types of data in different variables.

In [5]:

```
roll_number = 10
marks = 89.5
school = "ABC School"

print(roll_number)
print(marks)
print(school)
```

2. Printing Multiple Values

The `print()` function can display multiple values together.

In [6]:

```
print("Name:", name)
print("Age:", age)
print("Height:", height)
```

3. Input Function in Python

The `input()` function is used to take input from the user.

Important: By default, `input()` always takes input as a string.

Basic Input Example

In [7]:

```
user_name = input("Enter your name: ")
print("Hello", user_name)
```

Input Without Message

In [8]:

```
city = input()
print("City:", city)
```

Input is Always a String

Even if the user enters a number, Python treats it as text.

In [9]:

```
number = input("Enter a number: ")
print(number)
print(type(number))
```

Converting Input to Integer (int)

Used when you want whole numbers.

In [10]:

```
age_input = int(input("Enter your age: "))
```

```
print(age_input)
print(type(age_input))
```

Converting Input to Float (float)

Used for decimal values.

In [11]:

```
price = float(input("Enter price: "))
print(price)
print(type(price))
```

Taking Multiple Inputs (One by One)

In [12]:

```
first_name = input("Enter first name: ")
last_name = input("Enter last name: ")

print("Full Name:", first_name, last_name)
```

Summary

- Variables store data
- Python automatically detects variable types
- `print()` is used to display output
- `input()` takes user input as a string
- Type conversion is required for numbers